Results Overview

• Designs generated: 1

• Designs passing all filters: 1

• MMCIF files of final designs are in:

results/final_ranked_designs/final_1_designs

• Metrics and sequences of 1 final designs are in:

results/final_ranked_designs/final_designs_metrics_1.csv

• Metrics of all designs are in:

results/final_ranked_designs/all_designs_metrics.csv

You can rerun filtering (very quick), using this command with changed parameters:

```
-- boltzgen run input_spec.yaml --steps filtering --config filtering budget=60 alpha=0.05
```

You can also rerun filtering in a jupyter notebook if you want using `filter.ipynb`

What was run to produce this in the Filter task:

1. Filtering: each design is evaluated against mandatory thresholds.
2. Ranking: for every metric we compute its rank, then scale it by the metric's inverse-importance weight.

Designs with fewer passed filters are automatically penalised because the ranking key is the pair (num_filters_passed, metric). The overall quality score is the worst (maximum) of these scaled ranks.

The 10 best designs form the Top set.

3. Diversity: a lazy-greedy algorithm selects 1 designs that jointly maximise quality and minimise sequence similarity (sequence-identity distance). The trade-off is controlled by α = 0.001:

• α = 0 → 100 % quality focus (same as Top set)

• α = 1 → 100 % diversity focus (ignores quality)

• Quality – composite of metrics such as ESMFold2 ipSAE, salt-bridges, ΔSASA, etc. Each metric has an "inverse importance" weight (see "Sorting Criteria" table). A larger weight divides the rank by a bigger number and therefore down-weights that metric.

• Diversity – 1 − sequence identity between designs.

We use α = 0.001, meaning 100 % emphasis on quality, 0 % on diversity.

CSV Column Reference

Column	Description
id	filename to retrieve the file
designed_sequence	designed protein residues, with : separating chains
esmfold2_ipsae_min	ESMFold2 2021: minimum directional ipSAE, PAE <10 Å; best of five samples by default (higher = better)
esmfold2_design_to_target_ipsae	Directional ipSAE, designed polymer toward the selected target
esmfold2_target_to_design_ipsae	Directional ipSAE, selected target toward the designed polymer
designed_chain_sequence	full protein sequences of chains containing designed residues; : separates chains (synthesize separately)
num_design	number of designed residues
secondary_rank	intermediate rank from the sorting procedure
design_ptm	predicted TM score for intra-design contacts (higher = better)
design_iptm	predicted TM score for design–target contacts (higher = better)
design_to_target_iptm	same as design_iptm but for multi-chain designs
design_ipsae_min	min interaction pSAE: PAE-based confidence score for interaction (higher = better)
design_to_target_ipsae	interaction pSAE between design and target (higher = better)
min_design_to_target_pae	minimum PAE between design & target (lower = better)
plip_saltbridge	number of salt-bridge interactions
plip_hbonds	number of hydrogen-bond interactions
plip_hydrophobic	number of hydrophobic interactions
delta_sasa_original	ΔSASA when binder present vs absent
delta_sasa_refolded	same as above but on the refolded structure

Filtering & Sorting Criteria

This section first lists threshold filters that every design must pass (Filtering Criteria table)then explains how designs are ranked (Sorting Criteria table) weighted by their inverse importance.

Filtering Criteria

feature	lower_is_better	threshold	Pass
has_x	True	0.0	1
filter_rmsd	True	2.5	1
filter_rmsd_design	True	2.5	1
designfolding-filter_rmsd	True	2.5	1
ALA_fraction	True	0.3	1
GLY_fraction	True	0.3	1
GLU_fraction	True	0.3	1
LEU_fraction	True	0.3	1
VAL_fraction	True	0.3	1

Sorting Criteria

Metric	Inverse Importance
design_ptm	1
plip_hbonds_refolded	2
plip_saltbridge_refolded	2
delta_sasa_refolded	2
esmfold2_ipsae_min	1

Results Summary

Aggregate statistics

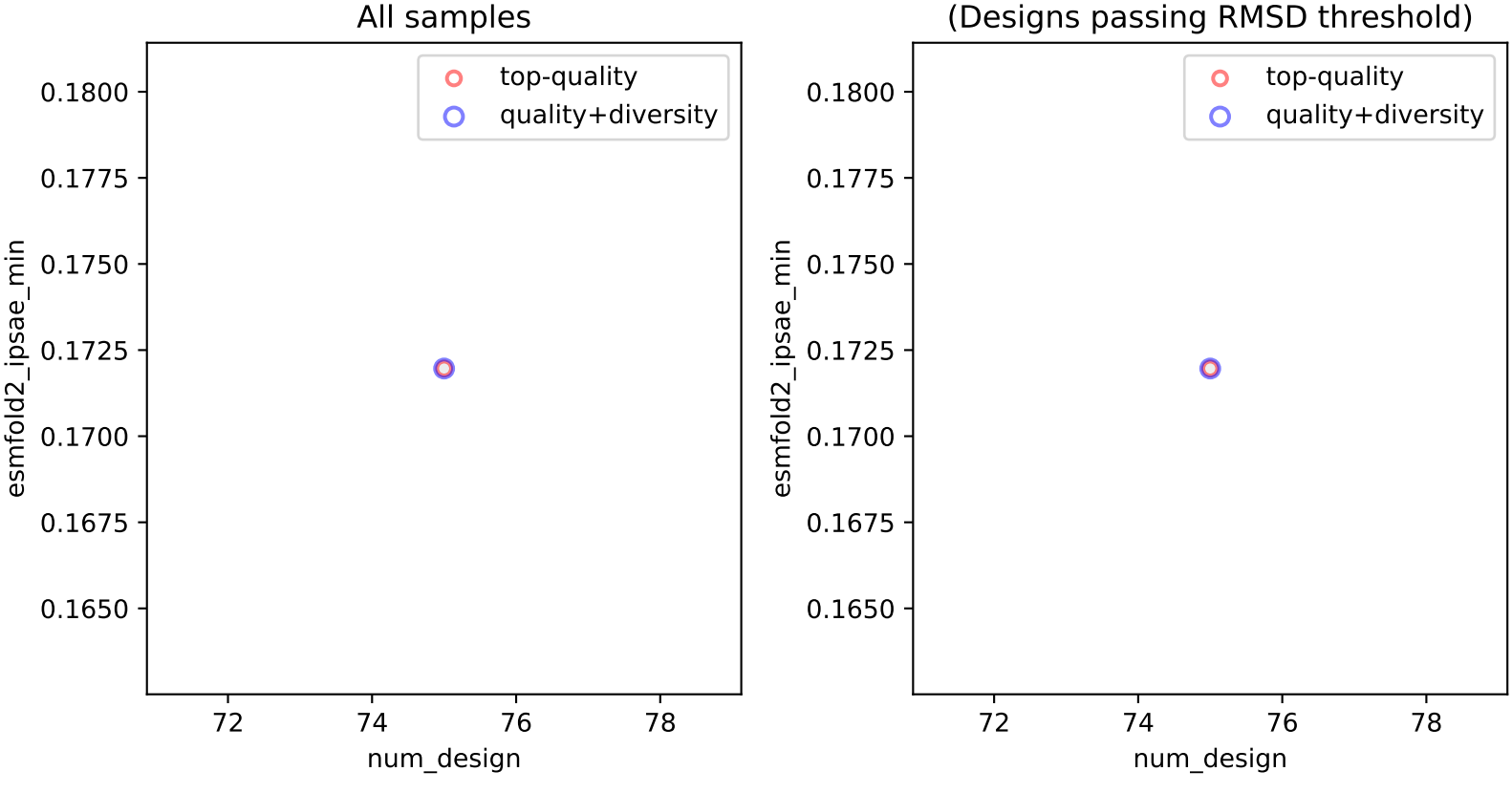
Quick numeric overview comparing (i) ALL incoming designs, (ii) the top-10 highest-quality designs, and (iii) the 1 quality+diversity designs produced by the lazy-greedy selection.

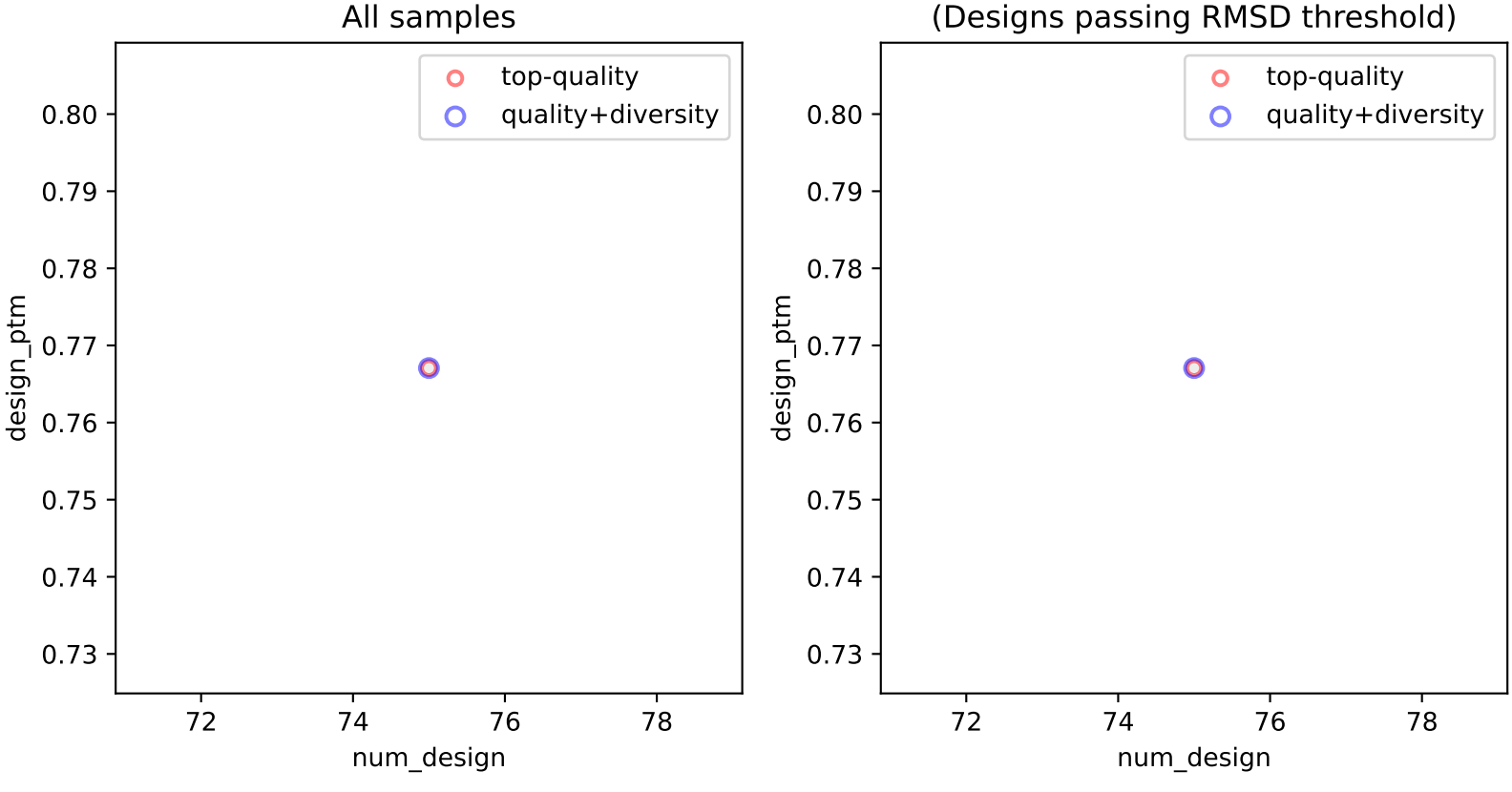
Metric	Mean	Mean top 10	Mean top 1 diverse
Num designs	1	-	-
esmfold2_ipsae_min	0.172	0.172	0.172
design_ptm	0.767	0.767	0.767
num_design	75.000	75.000	75.000
filter_rmsd	0.831	0.831	0.831
designfolding-filter_rmsd	1.332	1.332	1.332
design_ipsae_min	0.090	0.090	0.090
design_to_target_ipsae	0.148	0.148	0.148
delta_sasa_refolded	709.863	709.863	709.863
plip_saltbridge_refolded	1.000	1.000	1.000
plip_hbonds_refolded	6.000	6.000	6.000

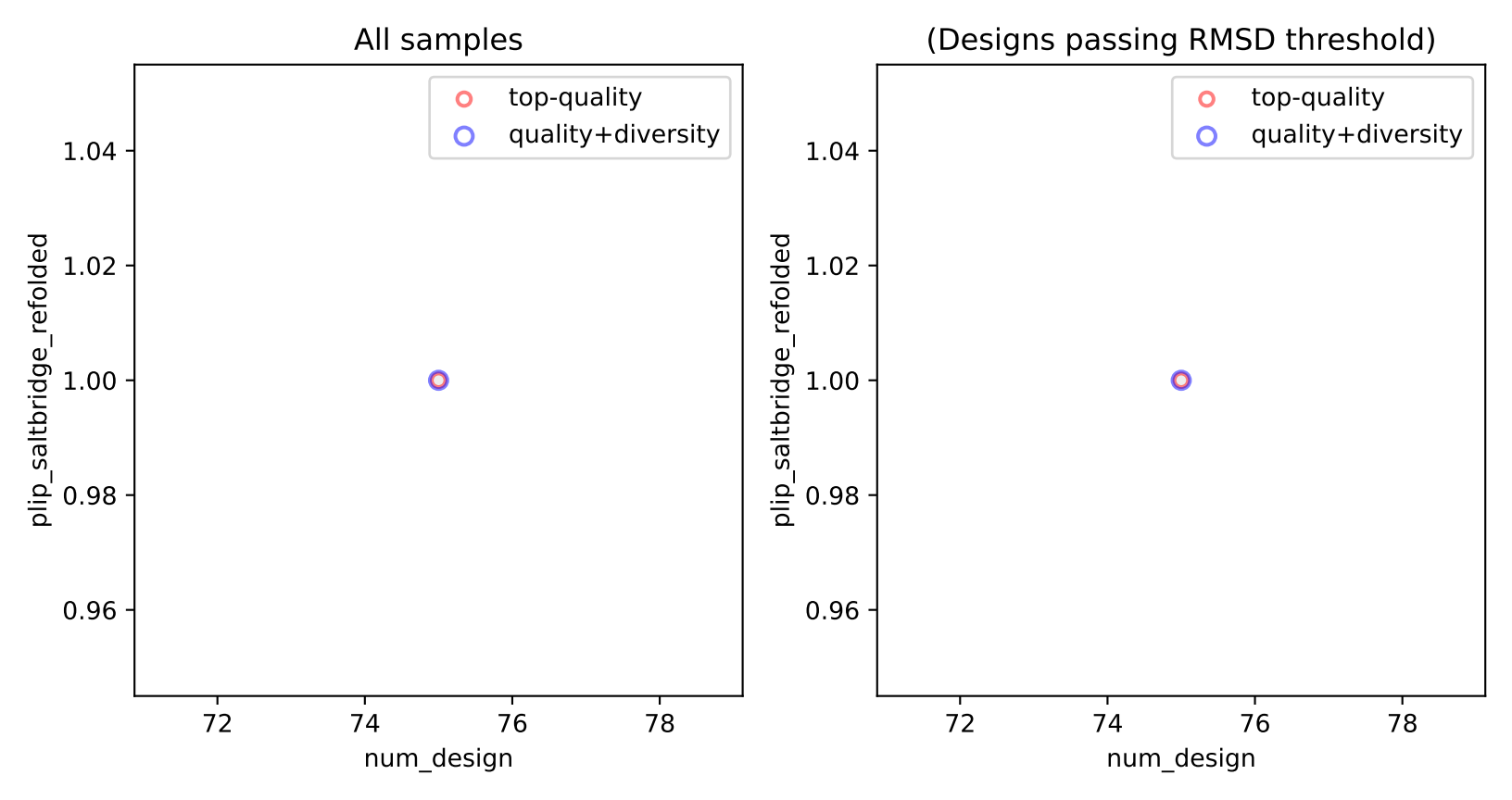
Scatter Plots

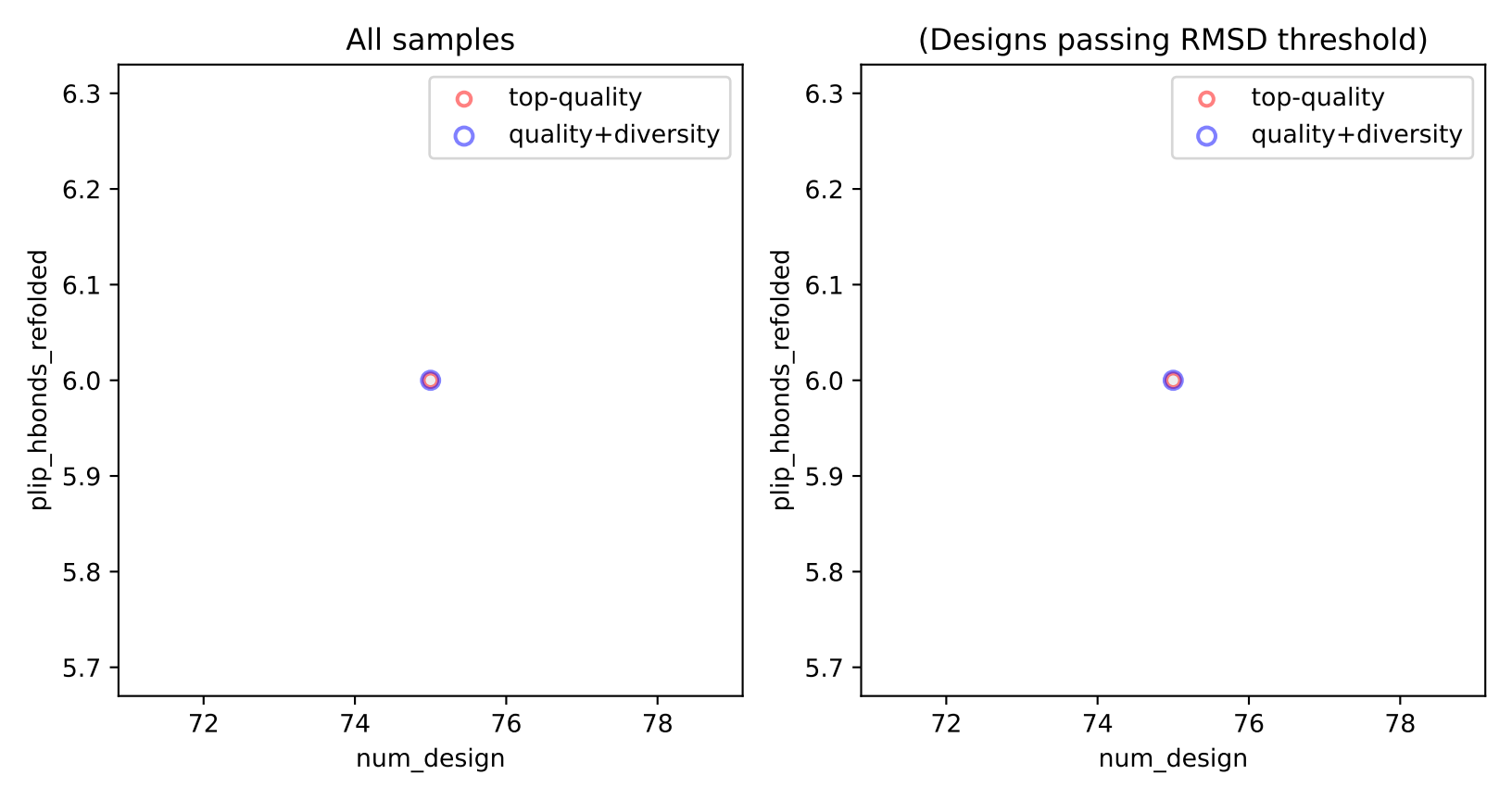
Metric relationships

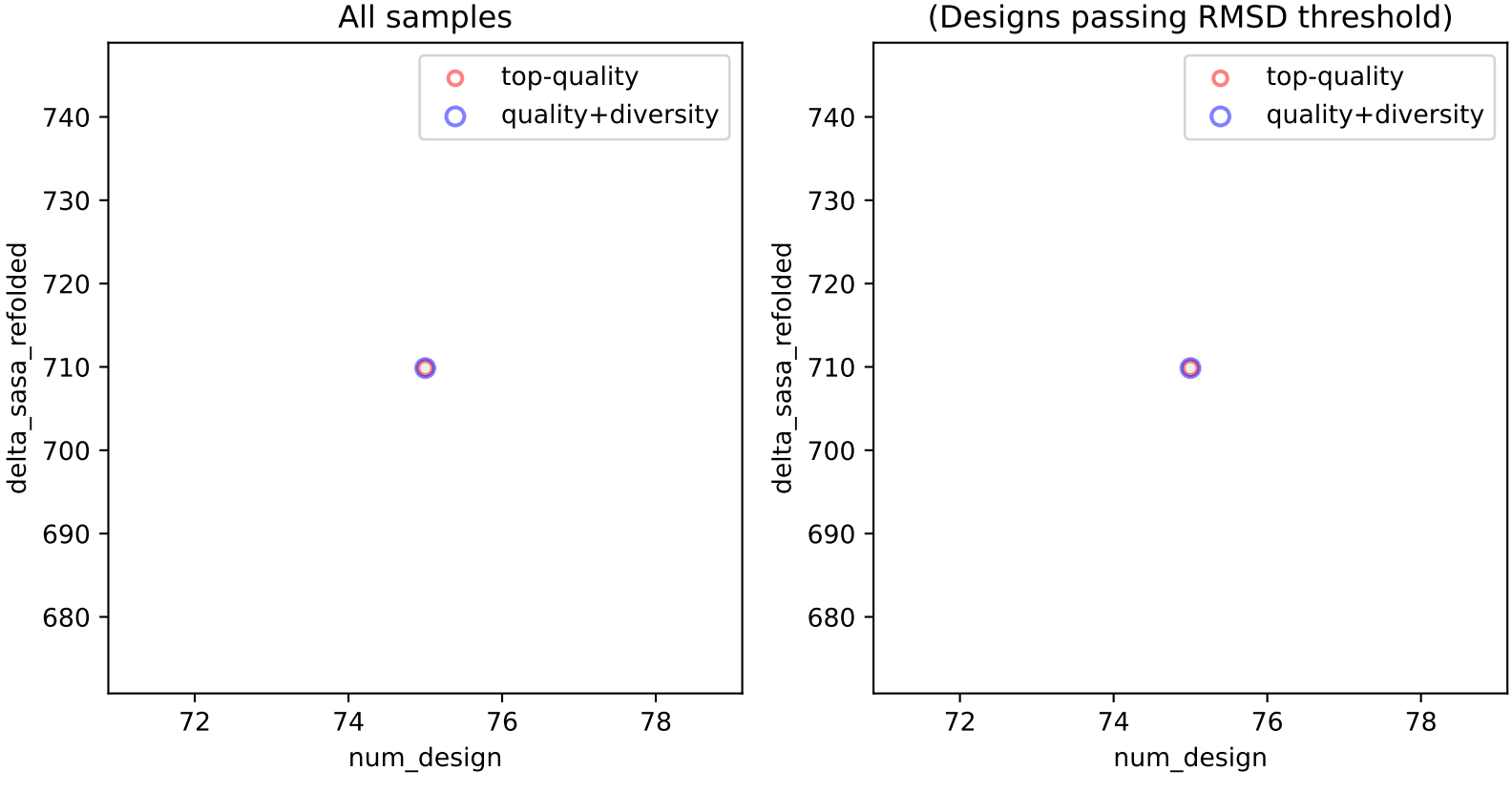
Each scatter page contains two panels:
• Left – all designs (grey) with overlays of Top (red) and Diverse (blue).
• Right – same but limited to designs passing the RMSD filter.

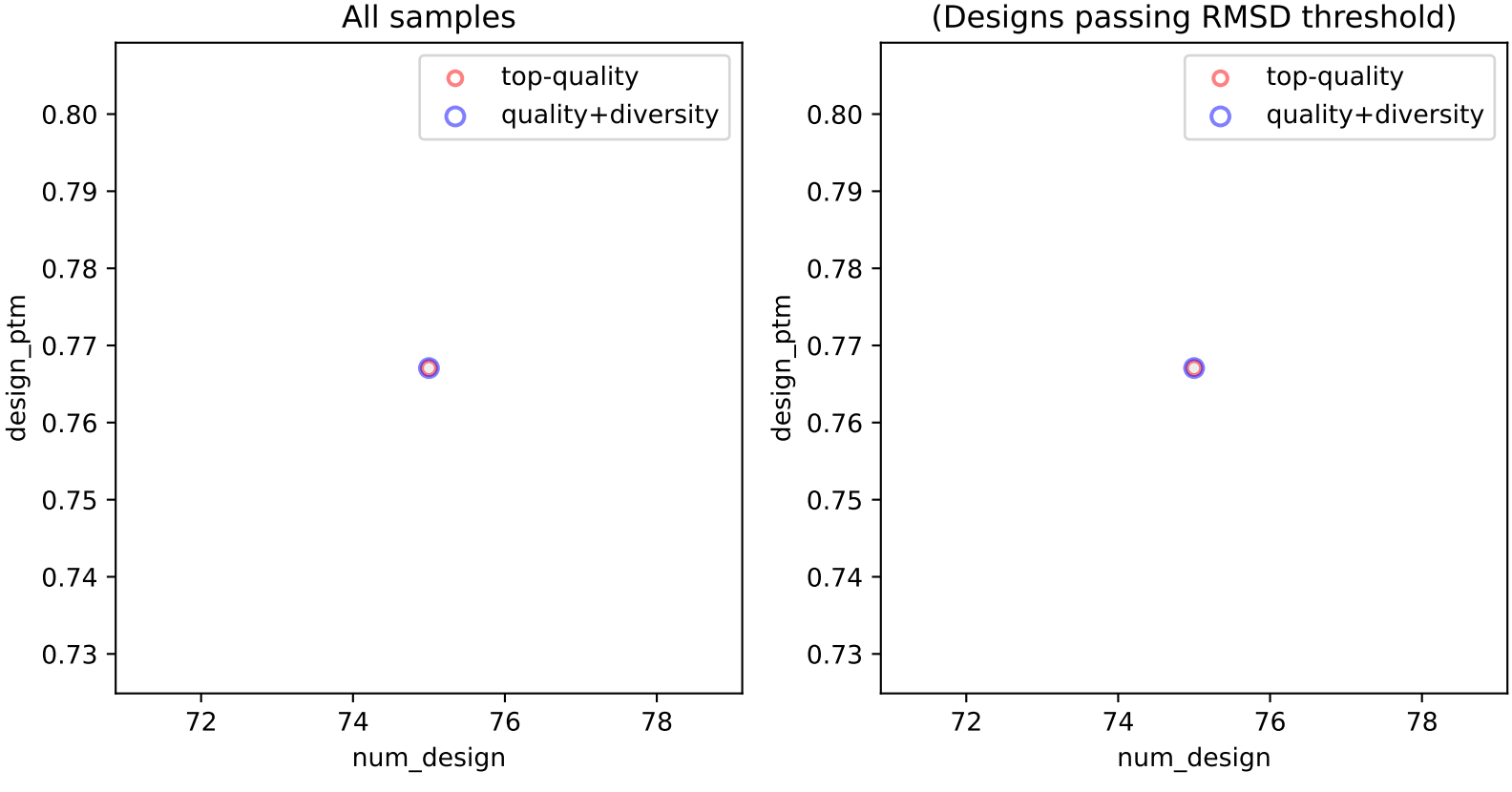

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
esmfold2_ipsae_min
0.1800
0.1775
0.1750
0.1725
0.1700
0.1675
0.1650
72
74
76
78
num_design

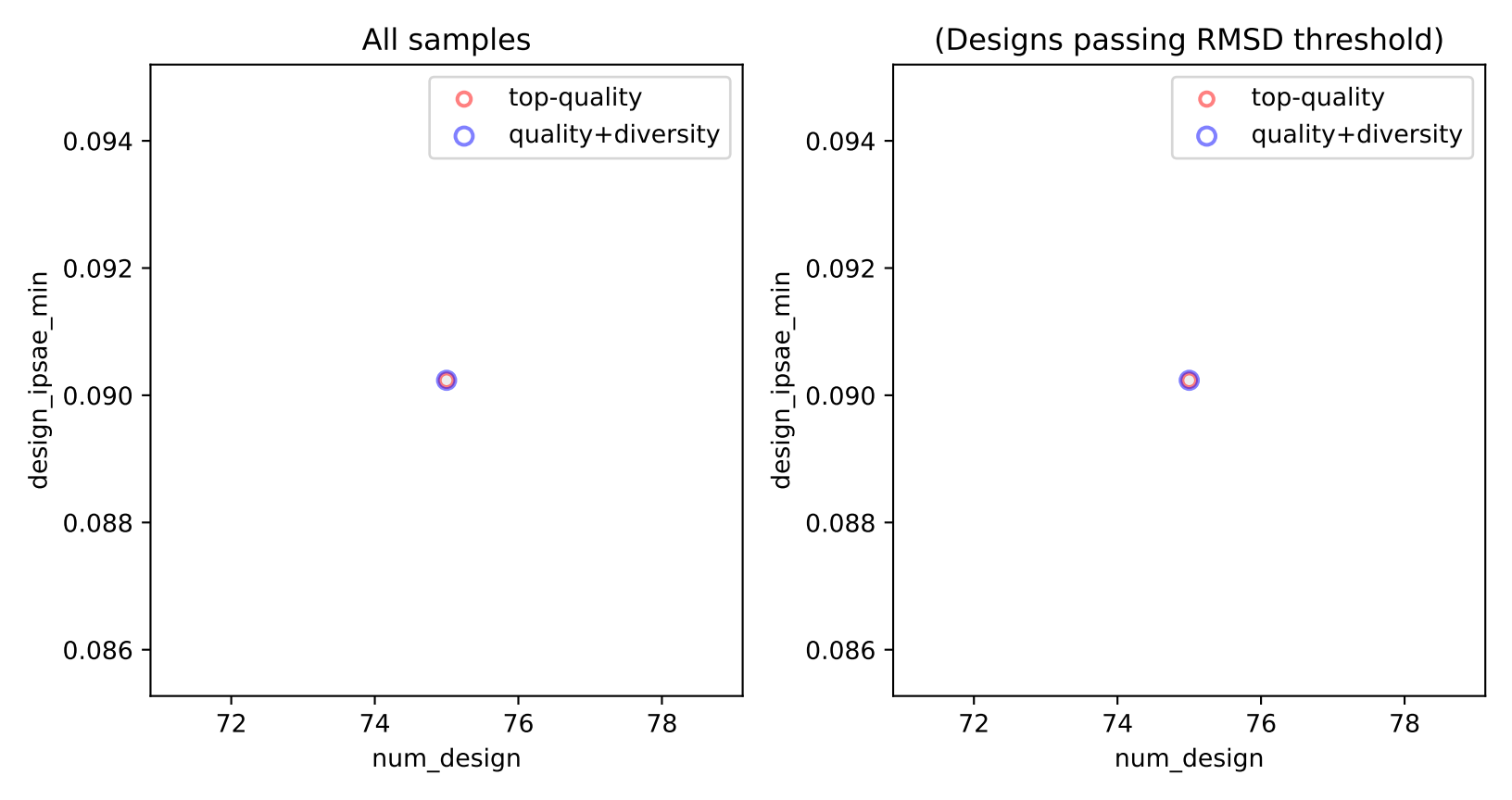

All samples
top-quality
quality+diversity
0.80
0.79
0.78
0.77
0.76
0.75
0.74
0.73
design_ptm
72
74
76
78
num_design
(Designs passing RMSD threshold)
top-quality
quality+diversity
0.80
0.79
0.78
0.77
0.76
0.75
0.74
0.73
design_ptm
72
74
76
78
num_design

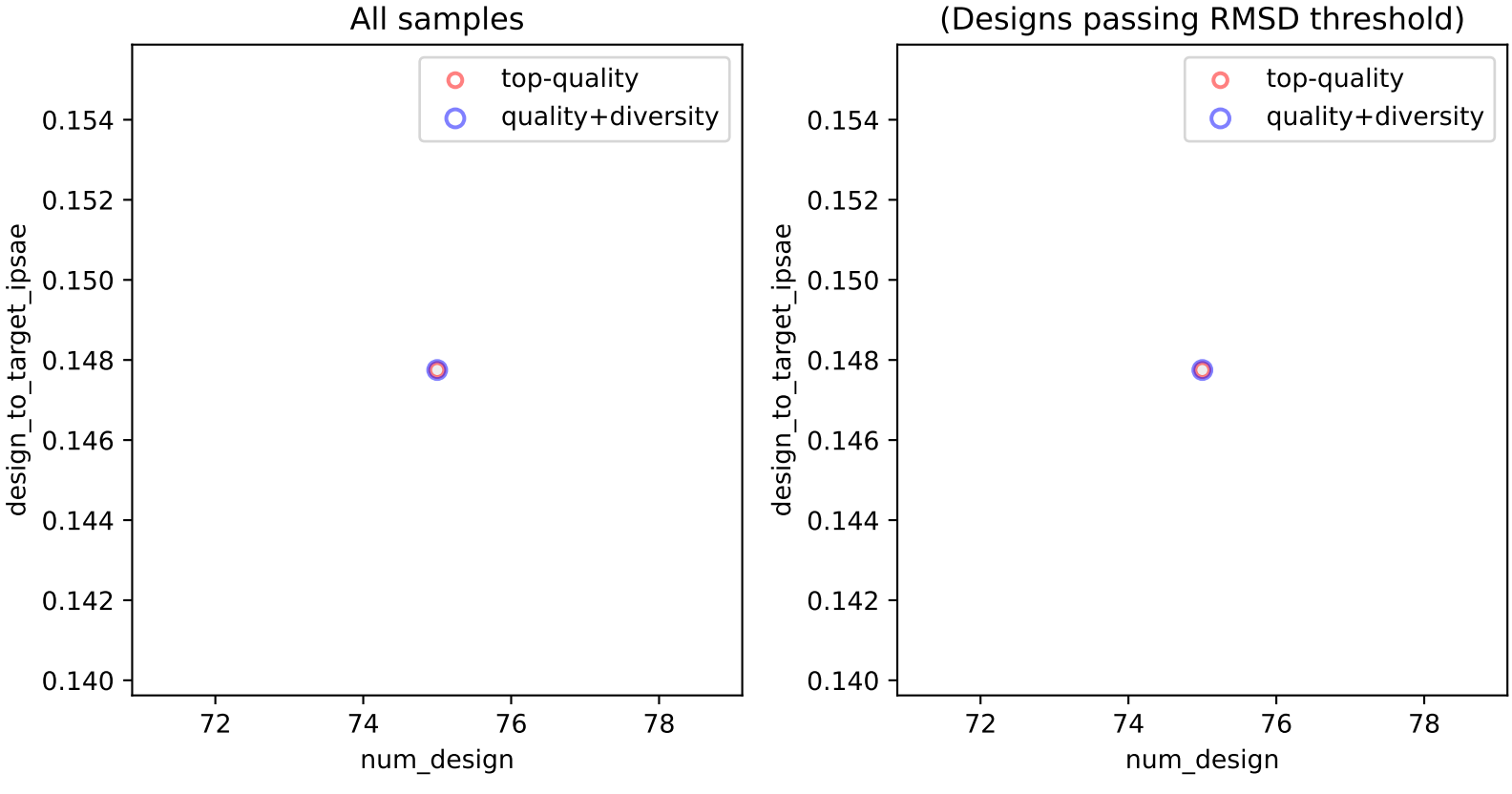

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
plip_saltbridge_refolded
num_design
1.04
1.02
1.00
0.98
0.96
72
74
76
78

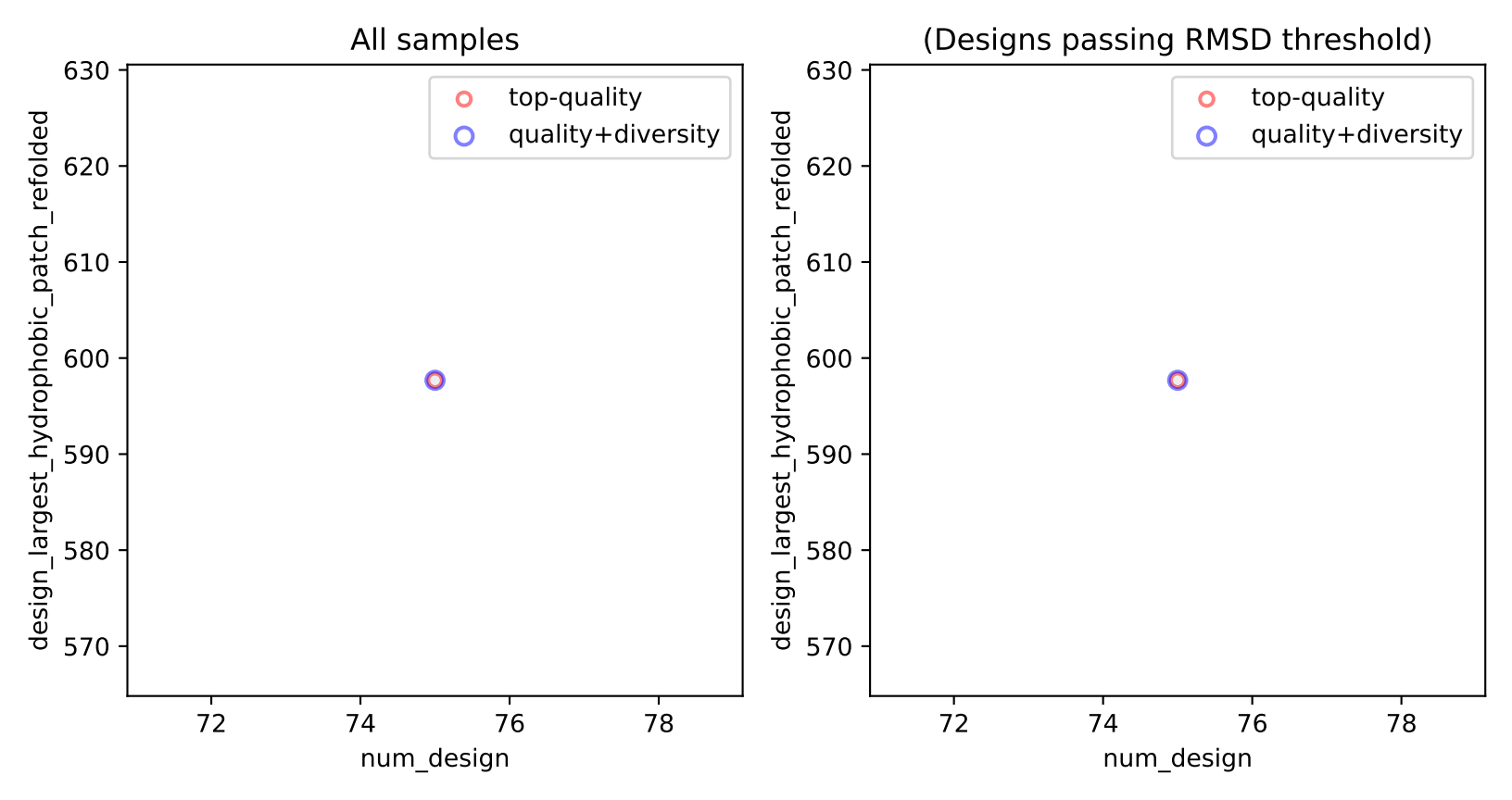

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
plip_hbonds_refolded
num_design
6.3
6.2
6.1
6.0
5.9
5.8
5.7
72
74
76
78

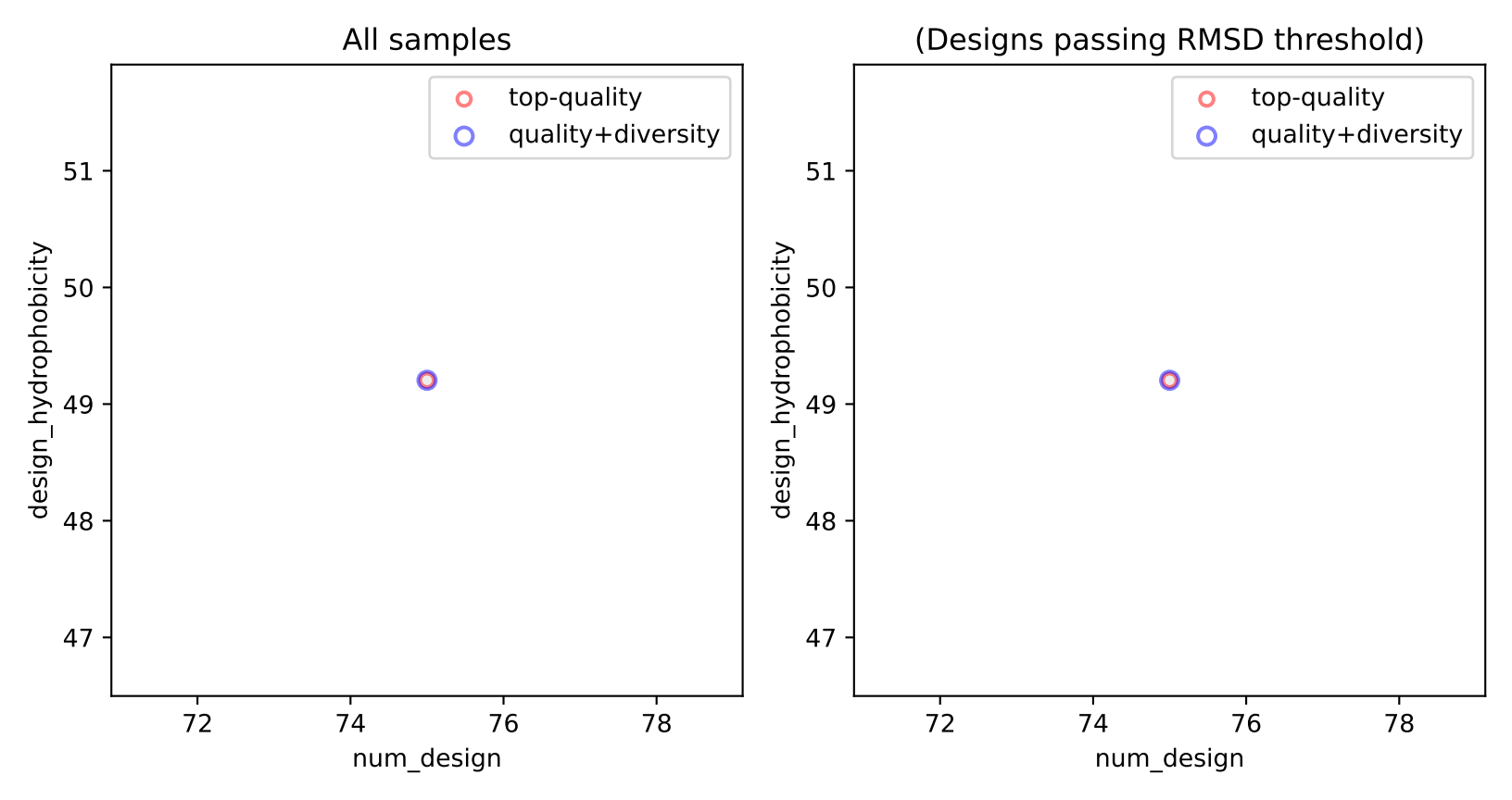

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
delta_sasa_refolded
num_design
740
730
720
710
700
690
680
72
74
76
78


All samples
top-quality
quality+diversity
0.80
0.79
0.78
0.77
0.76
0.75
0.74
0.73
design_ptm
72
74
76
78
num_design
(Designs passing RMSD threshold)
top-quality
quality+diversity
0.80
0.79
0.78
0.77
0.76
0.75
0.74
0.73
design_ptm
72
74
76
78
num_design


All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_ipsae_min
num_design
0.094
0.092
0.090
0.088
0.086
72
74
76
78


All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_to_target_ipsae
num_design
0.154
0.152
0.150
0.148
0.146
0.144
0.142
0.140
72
74
76
78

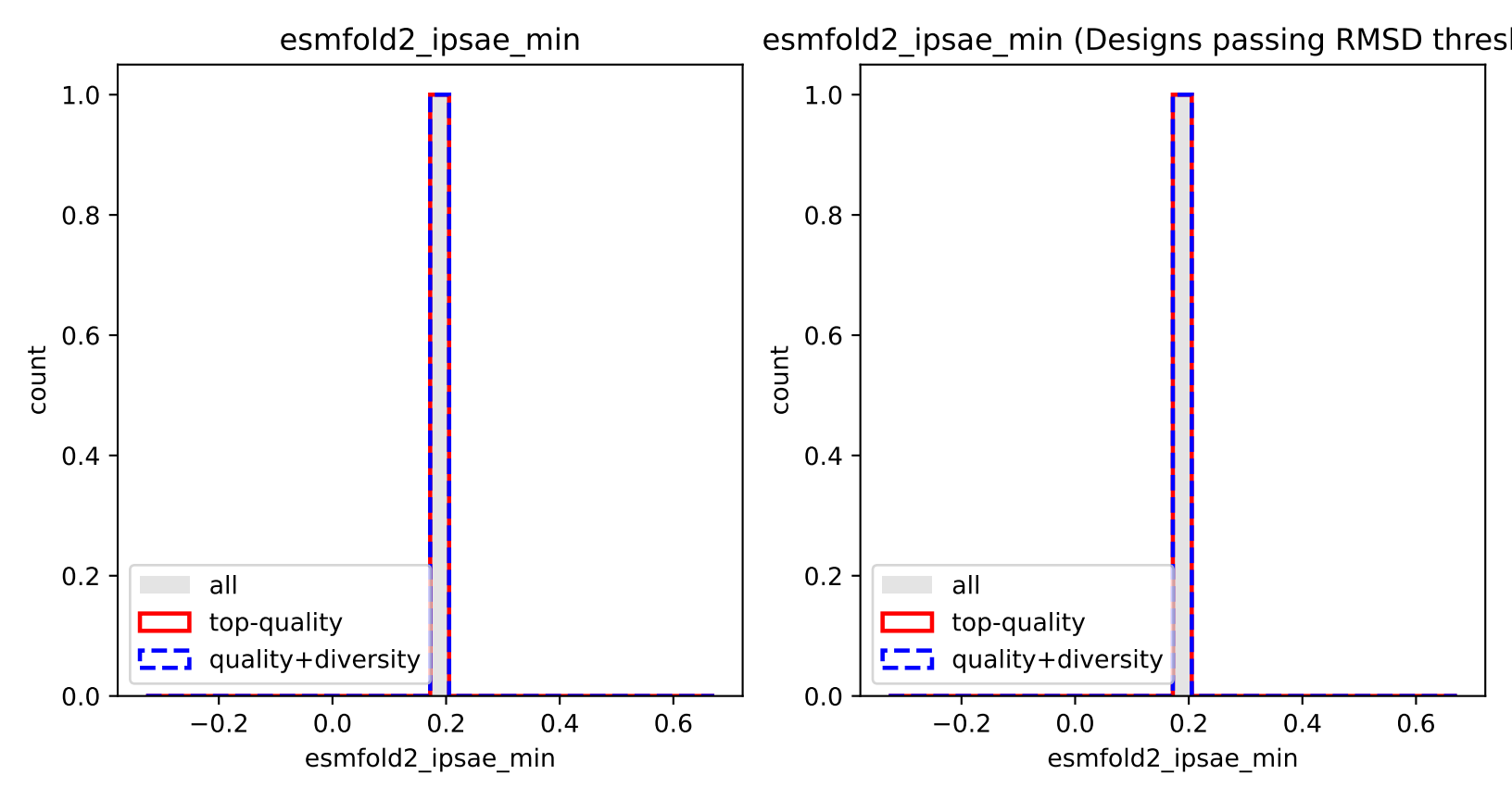

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_largest_hydrophobic_patch_refolded
num_design
630
620
610
600
590
580
570
72
74
76
78

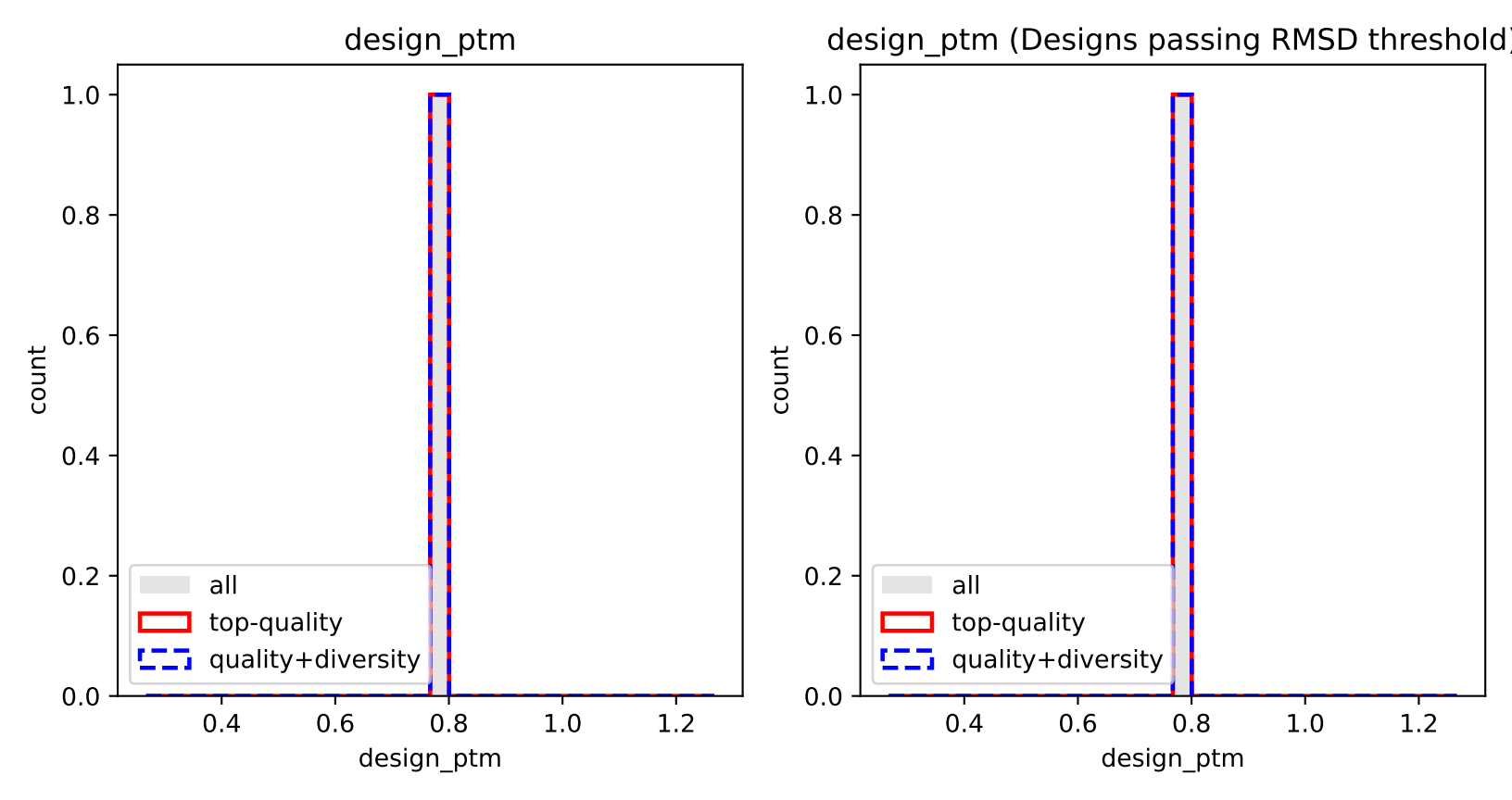

All samples
(Designs passing RMSD threshold)
top-quality
quality+diversity
design_hydrophobicity
num_design
47
48
49
50
51
72
74
76
78

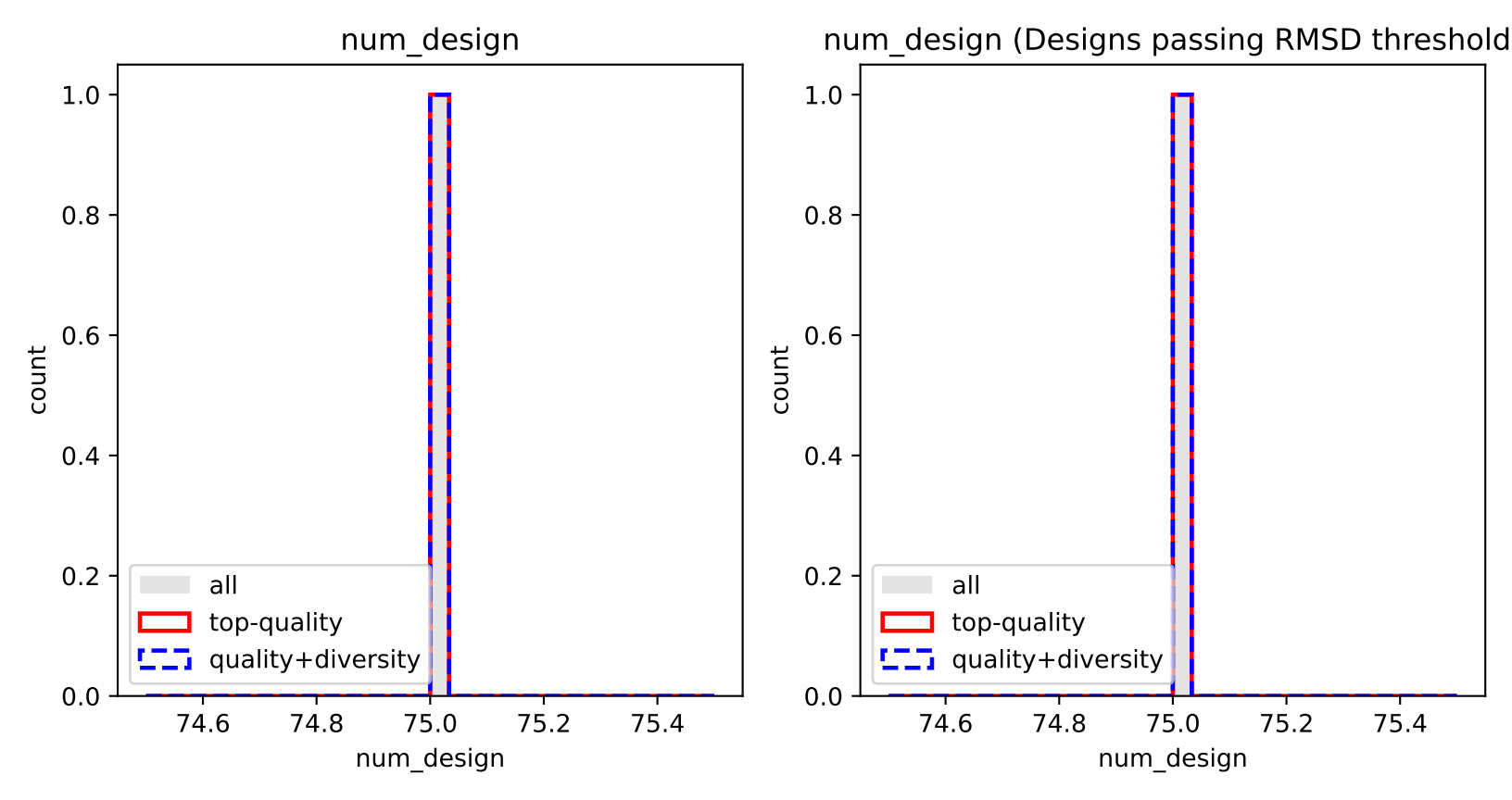
Metric Distributions

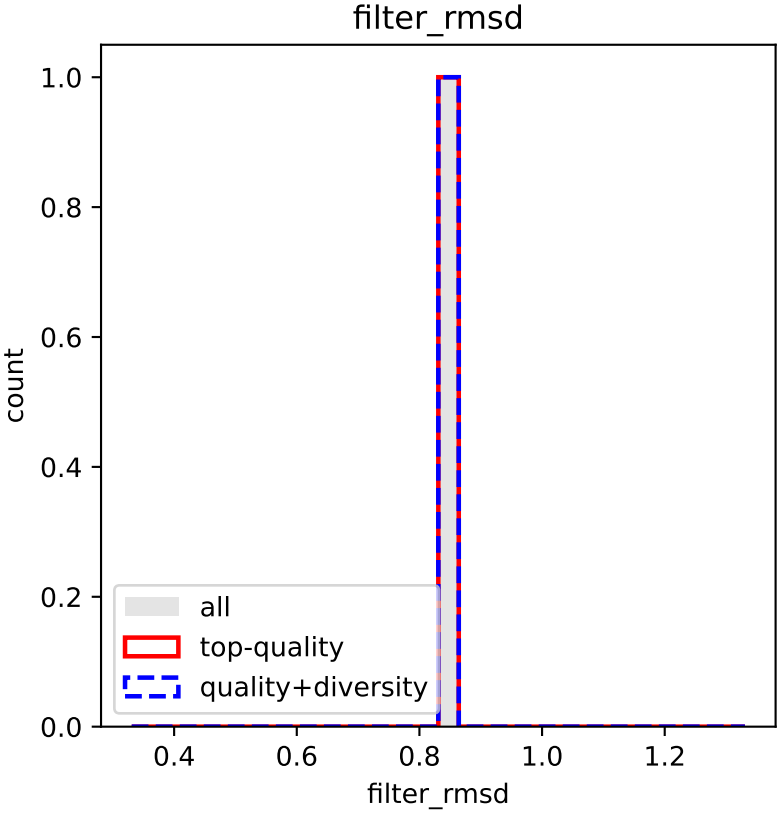
Histograms

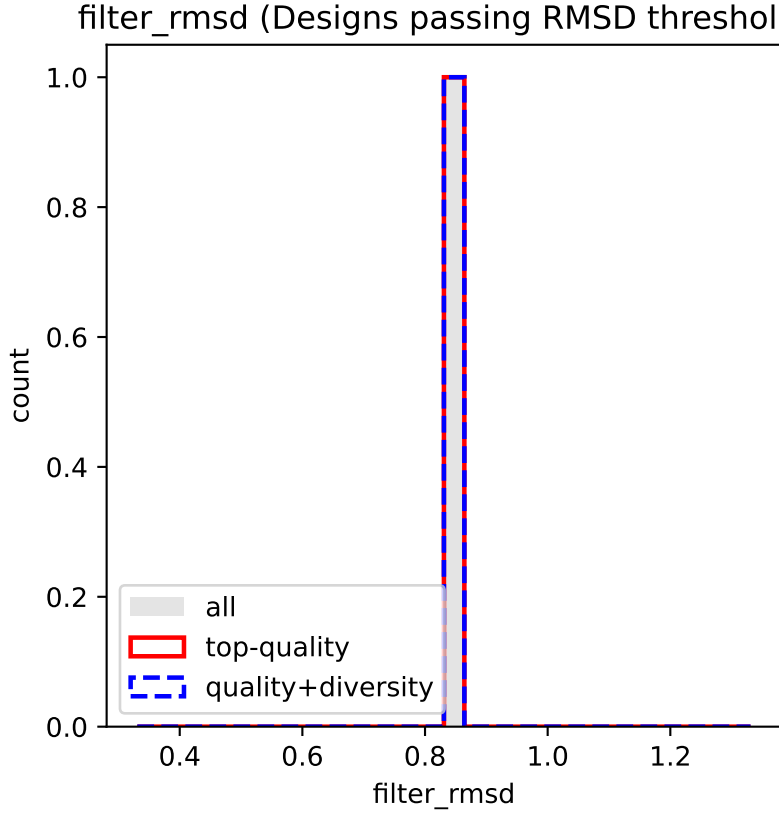
Distribution of each metric across all designs (grey) with overlays for Top-quality (red outline) and Diversity-optimised (blue dashed) subsets. The right panel repeats the histogram but only for designs that pass the RMSD threshold.

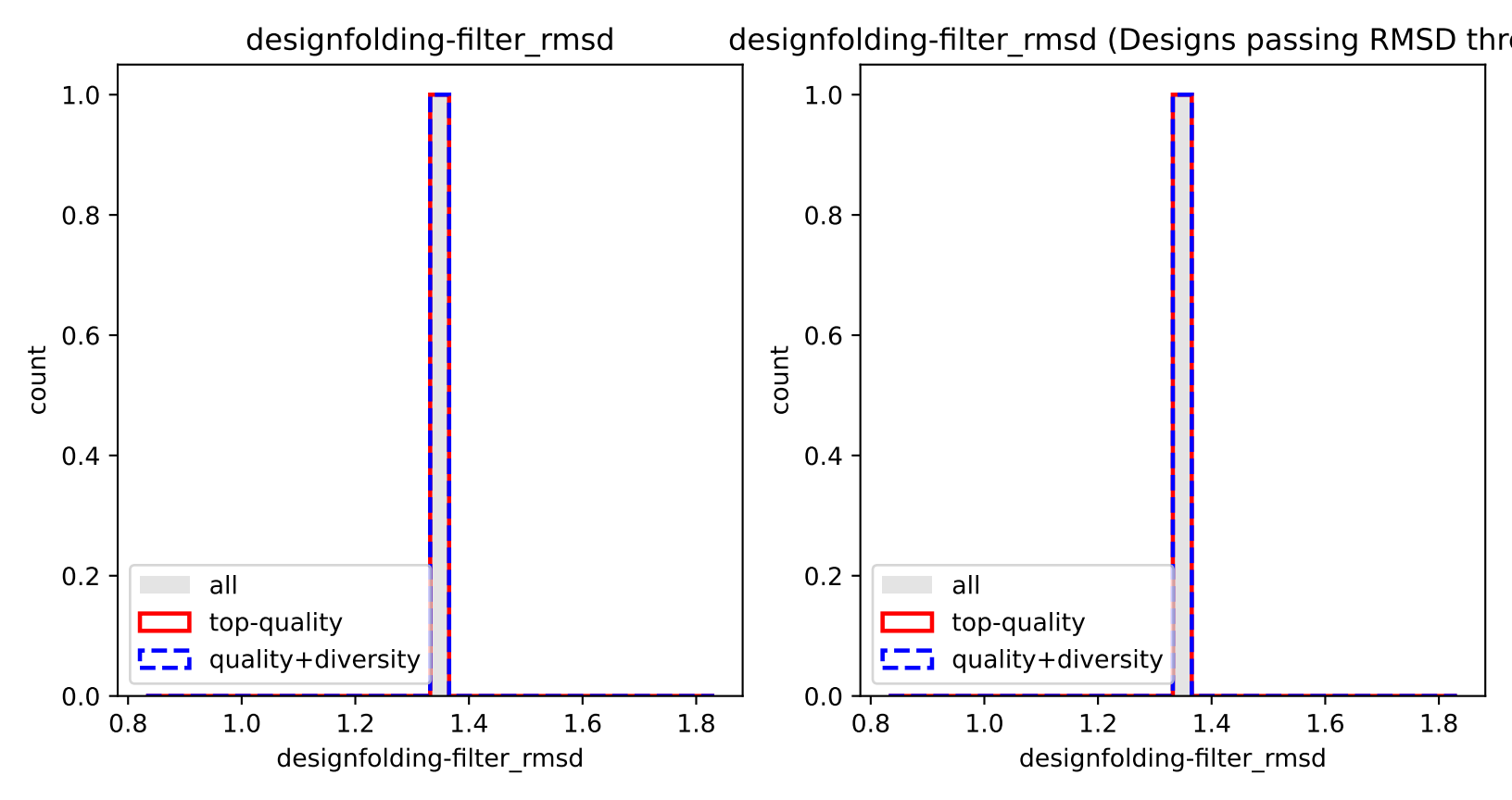

esmfold2_ipsae_min
esmfold2_ipsae_min (Designs passing RMSD thresh
count
esmfold2_ipsae_min
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
−0.2
0.0
0.2
0.4
0.6

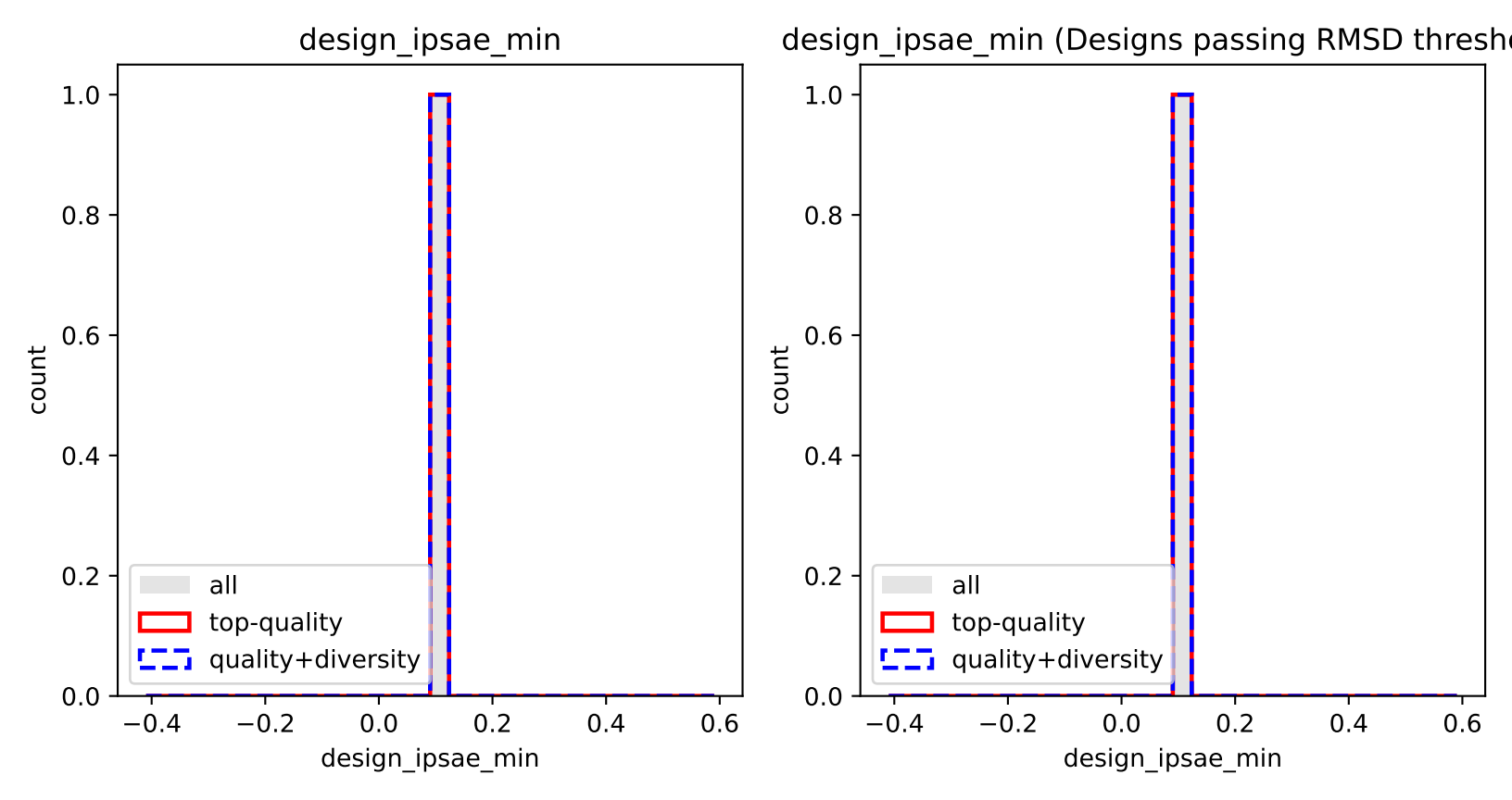

design_ptm
count
design_ptm
all
top-quality
quality+diversity
design_ptm (Designs passing RMSD threshold)
count
design_ptm
all
top-quality
quality+diversity

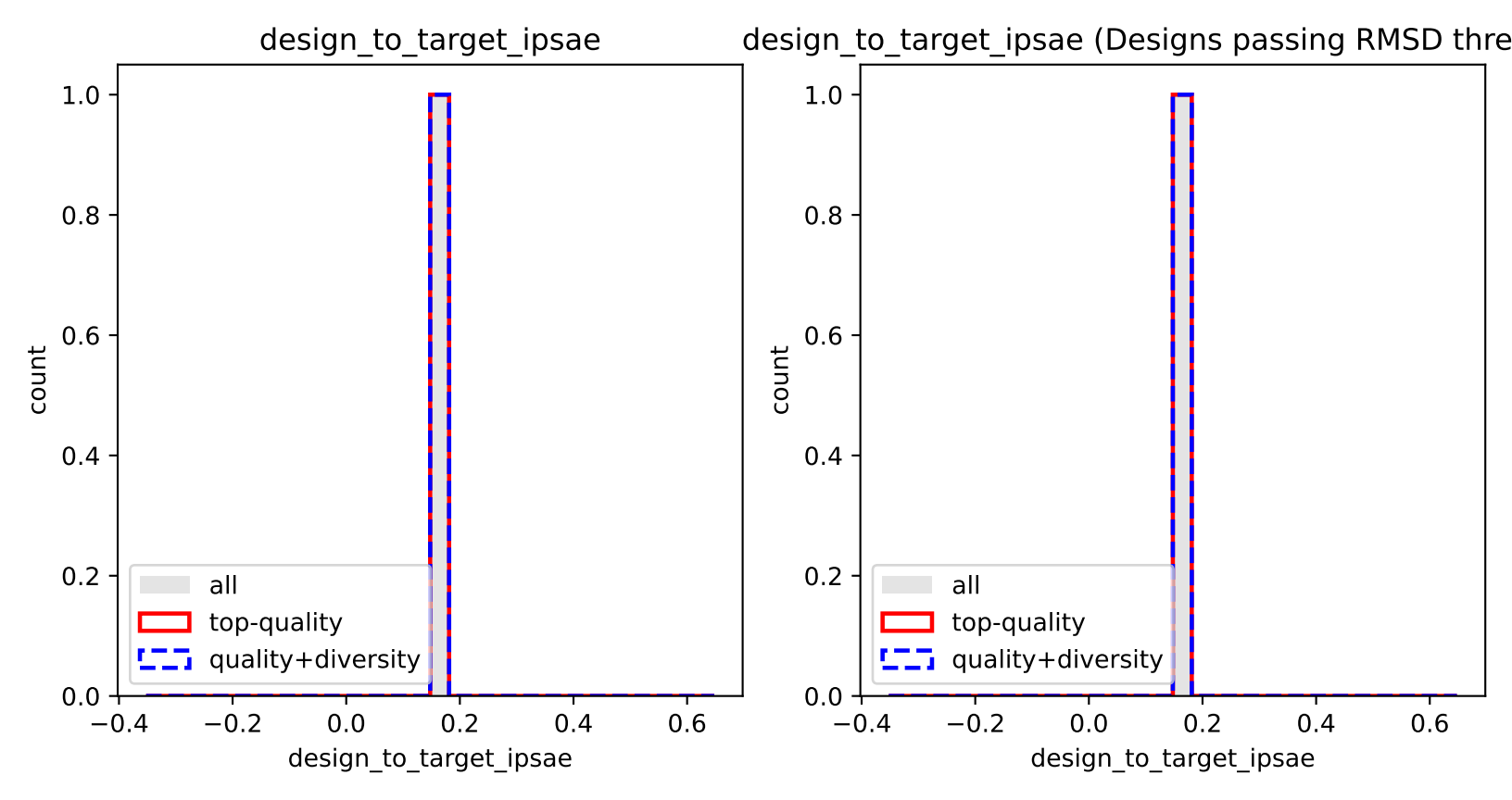

num_design
count
num_design
all
top-quality
quality+diversity
num_design (Designs passing RMSD threshold
count
num_design
all
top-quality
quality+diversity

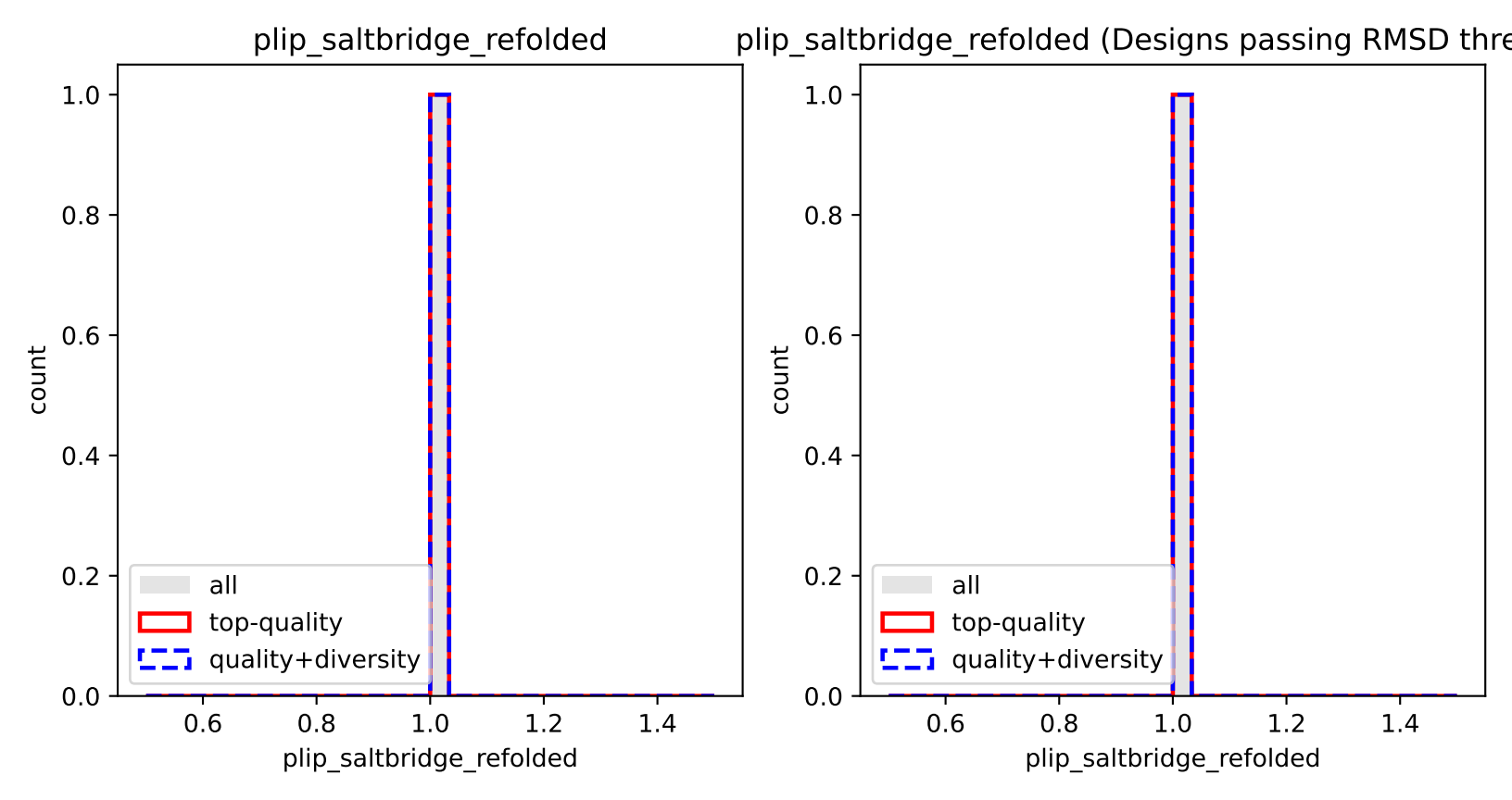

filter_rmsd
1.0
0.8
0.6
0.4
0.2
0.0
count
all
top-quality
quality+diversity
0.4
0.6
0.8
1.0
1.2
filter_rmsd

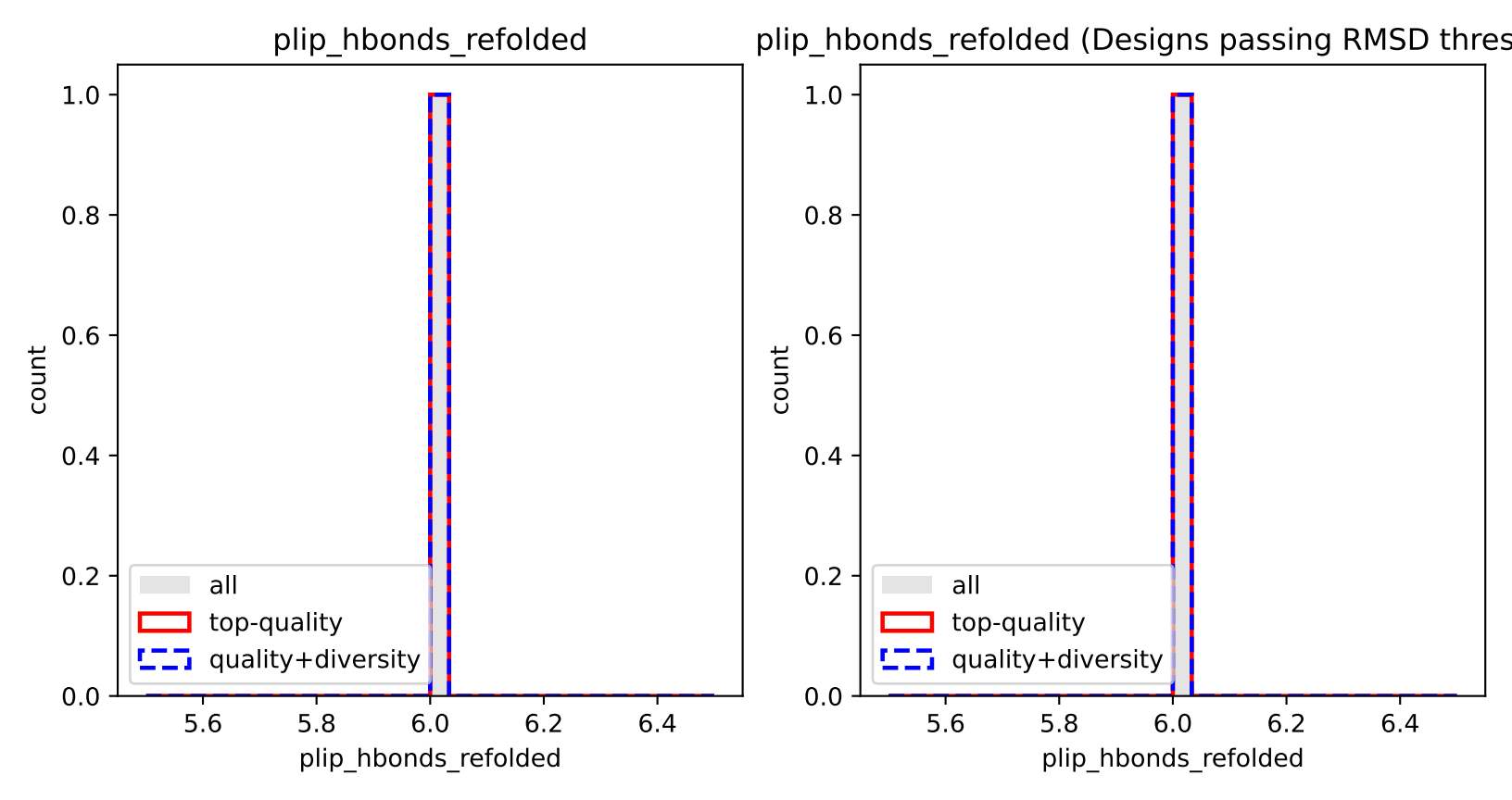

filter_rmsd (Designs passing RMSD threshold)
1.0
0.8
0.6
0.4
0.2
0.0
count
all
top-quality
quality+diversity
0.4
0.6
0.8
1.0
1.2
filter_rmsd

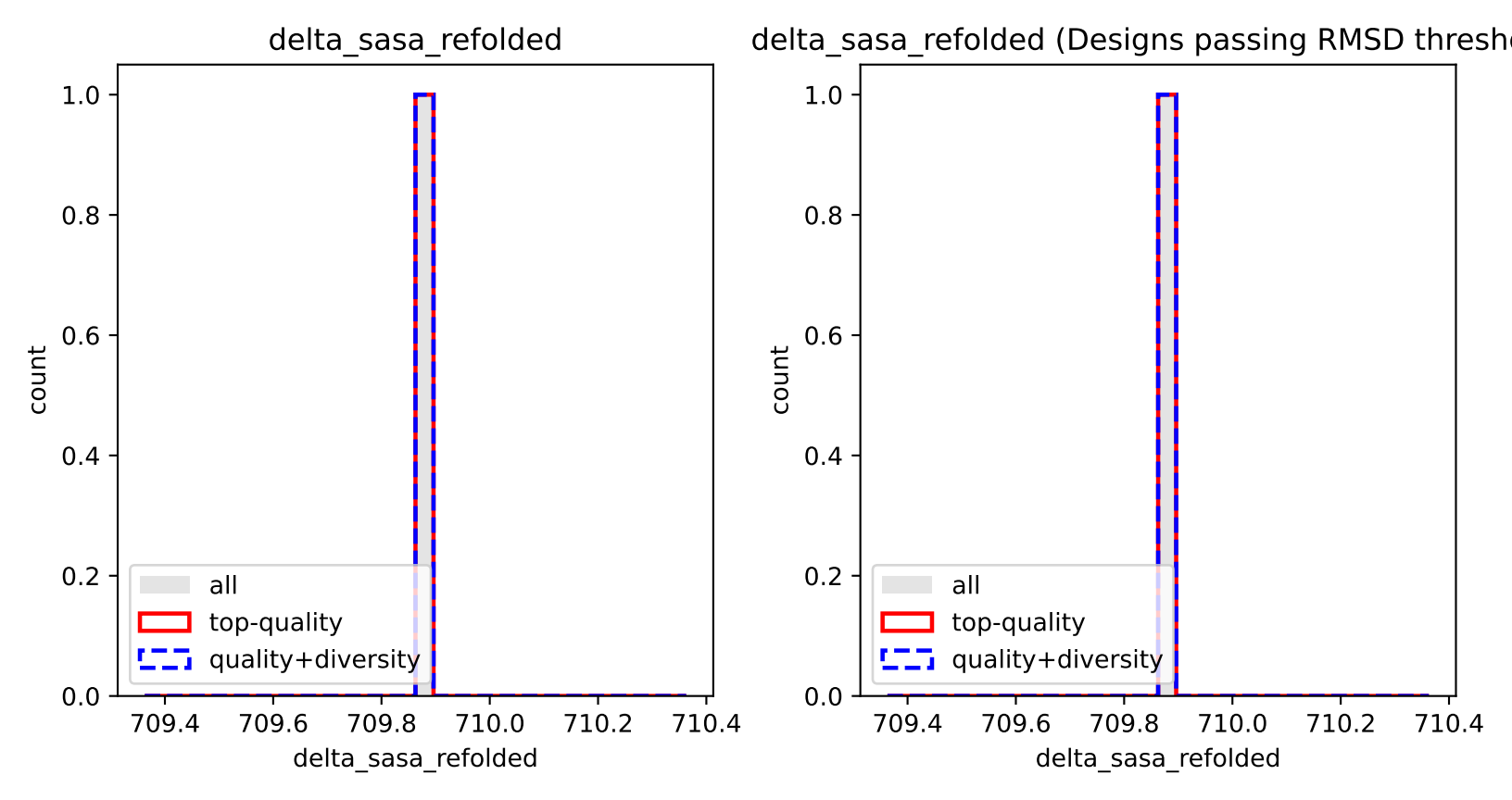

designfolding-filter_rmsd
designfolding-filter_rmsd (Designs passing RMSD thr
count
designfolding-filter_rmsd
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
0.8
1.0
1.2
1.4
1.6
1.8

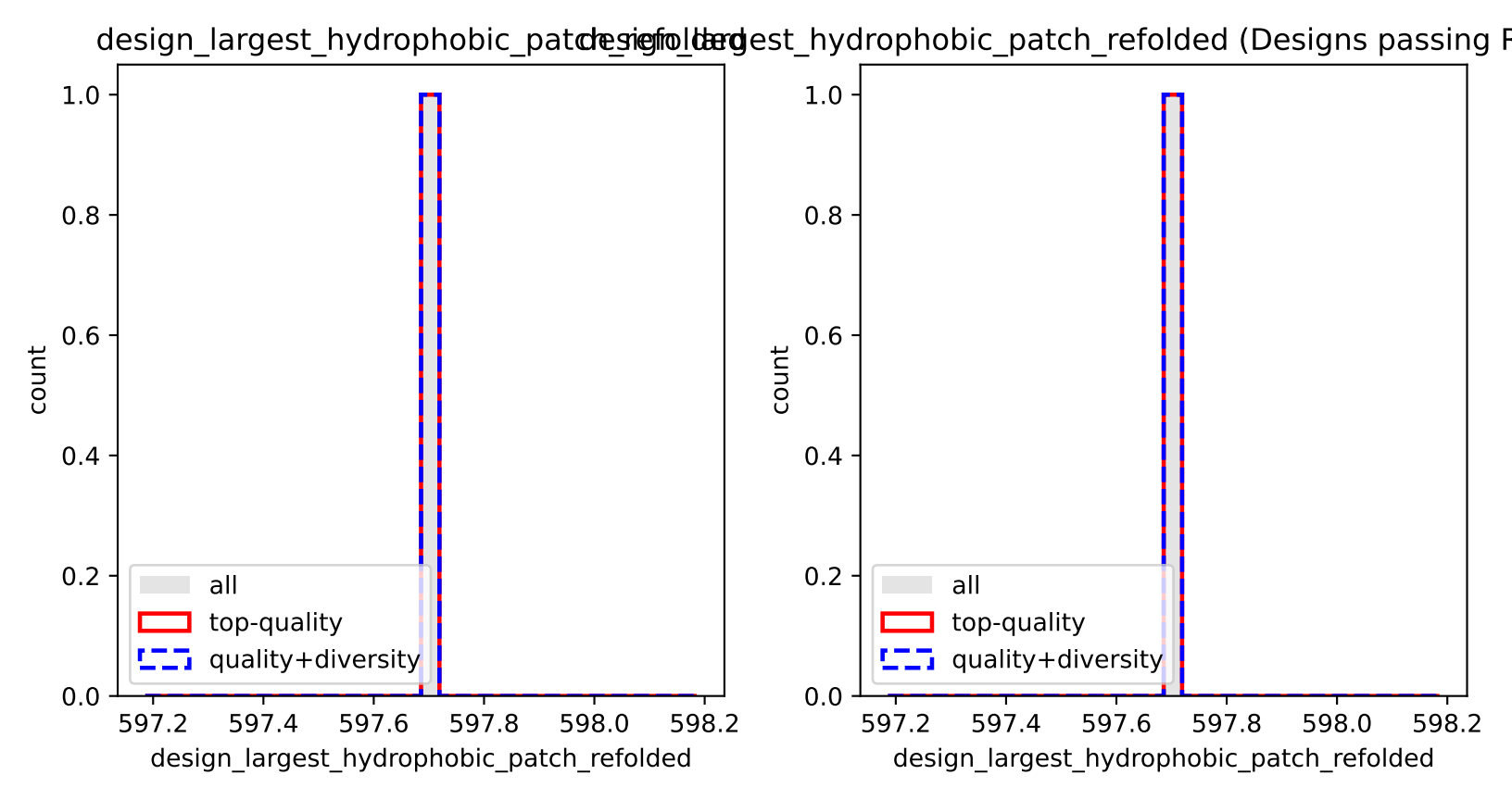

design_ipsae_min
design_ipsae_min (Designs passing RMSD thresh
count
design_ipsae_min
all
top-quality
quality+diversity
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6

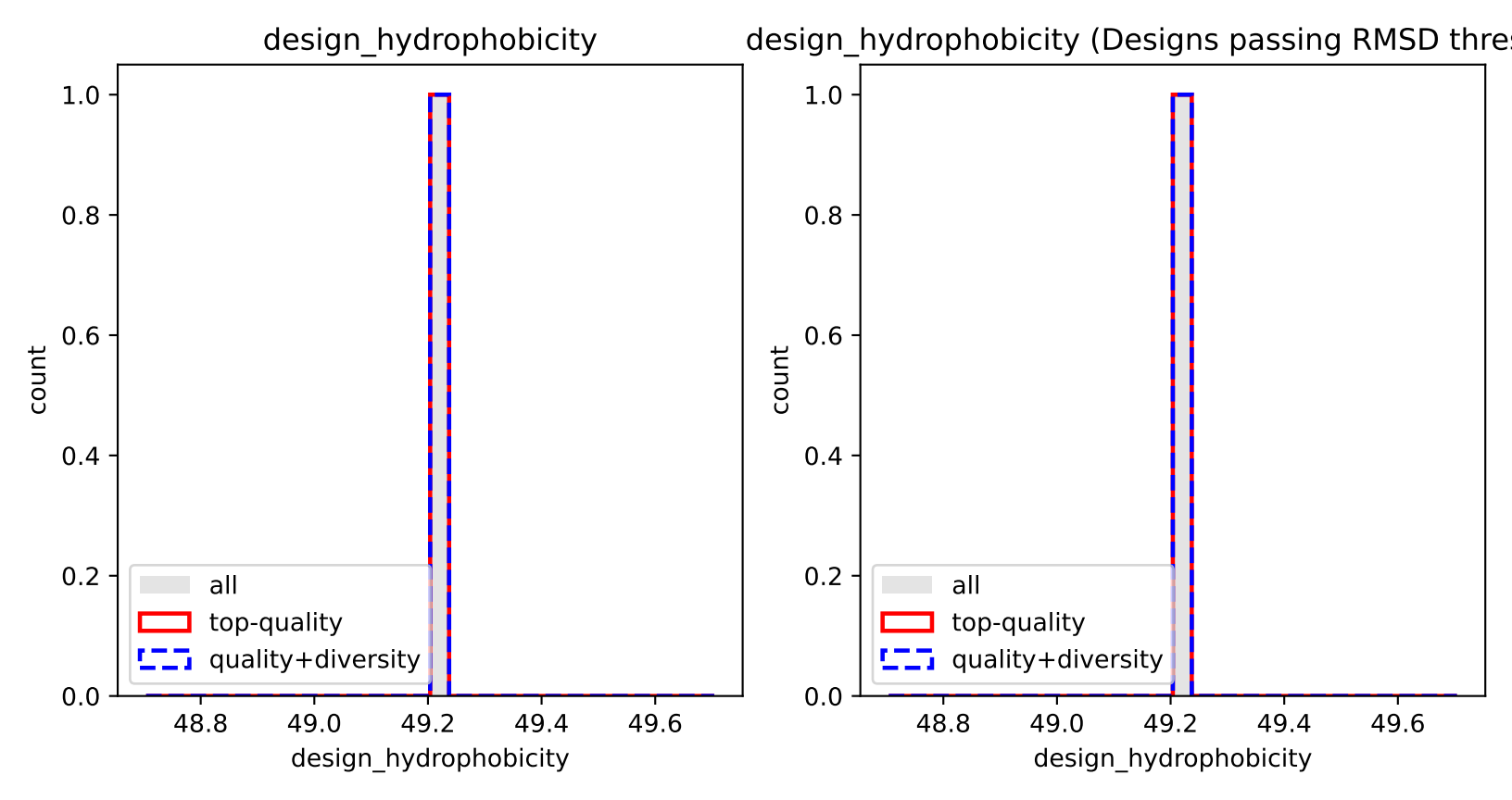

design_to_target_ipsae
design_to_target_ipsae (Designs passing RMSD thre
count
design_to_target_ipsae
all
top-quality
quality+diversity
1.0
0.8
0.6
0.4
0.2
0.0
−0.4
−0.2
0.0
0.2
0.4
0.6


plip_saltbridge_refolded
plip_saltbridge_refolded (Designs passing RMSD thre
count
plip_saltbridge_refolded
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
0.6
0.8
1.0
1.2
1.4


plip_hbonds_refolded
plip_hbonds_refolded (Designs passing RMSD thres
count
plip_hbonds_refolded
all
top-quality
quality+diversity
0.0
0.2
0.4
0.6
0.8
1.0
5.6
5.8
6.0
6.2
6.4


delta_sasa_refolded
delta_sasa_refolded (Designs passing RMSD thresh
count
1.0
0.8
0.6
0.4
0.2
0.0
709.4
709.6
709.8
710.0
710.2
710.4
delta_sasa_refolded
all
top-quality
quality+diversity


design_largest_hydrophobic_patch_refolded
design_largest_hydrophobic_patch_refolded (Designs passing F
count
1.0
0.8
0.6
0.4
0.2
0.0
all
top-quality
quality+diversity
597.2
597.4
597.6
597.8
598.0
598.2
design_largest_hydrophobic_patch_refolded


design_hydrophobicity
design_hydrophobicity (Designs passing RMSD thres
count
1.0
0.8
0.6
0.4
0.2
0.0
48.8
49.0
49.2
49.4
49.6
design_hydrophobicity
all
top-quality
quality+diversity